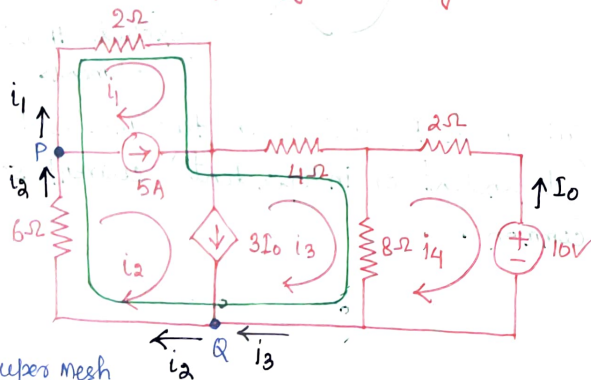


• Note the following properties of Supermesh

- The current source in a Supermesh provides the constraint equation necessary to solve for the mesh currents
- A Supermesh has no current of its own
- A Supermesh requires the application of both KVL and KCL.

Q.1: find i_1 to i_4 using mesh Analysis.



Super mesh

$$\Rightarrow 2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

(or)

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0$$

for the independent current source, we apply KCL at node P:

$$i_2 = i_1 + 5$$

for the dependent source, we apply KCL at node Q:

$$i_2 = i_3 + 3I_o$$

$$\text{But } I_o = -I_4$$

$$i_2 = i_3 - 3I_4$$

Apply KVL in mesh 4

$$\Rightarrow 2i_4 + 8(i_4 - i_3) + 10 = 0$$

(or)

$$5i_4 - 4i_3 = -5$$

$$\begin{cases} i_1 = -7.5 \text{ A} \\ i_2 = -2.5 \text{ A} \\ i_3 = 3.93 \text{ A} \\ i_4 = 2.143 \text{ A} \end{cases}$$

NODAL ANALYSIS

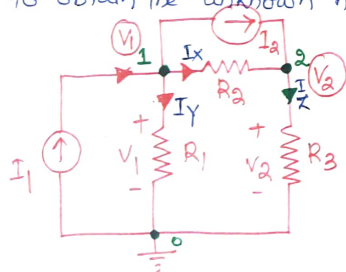
(*) Nodal Analysis provides a general procedure for analyzing CKTs using node voltages as the circuit variables.

(a) select a node as reference node. Assign voltages $V_1, V_2, \dots, V_{n-1}$ to the $(n-1)$ nodes. The voltages are referenced with respect to the reference node.

(b) Apply KCL to each of the $n-1$ non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.

(c) Solve the resulting simultaneous equations to obtain the unknown node voltages.

Q.



At Node-1

$$I_1 = I_2 + I_x + I_Y$$

$$\Rightarrow I_1 = I_2 + \frac{V_1 - V_2}{R_2} + \frac{V_1}{R_1}$$

OR

$$\Rightarrow I_1 = I_2 + (V_1 - V_2)G_2 + V_1G_1$$

At Node 2:

$$I_x + I_2 = I_2$$

$$I_2 + \frac{(V_1 - V_2)}{R_2} = \frac{V_2}{R_3} \quad (\text{OR}) \quad I_2 + G_2(V_1 - V_2) = G_3V_2$$

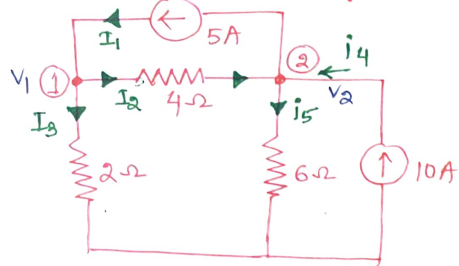
finally (OR)

$$I_1 = I_2 + G_1V_1 + G_2(V_1 - V_2)$$

$$I_2 + G_2(V_1 - V_2) = G_3V_2$$

$$\begin{bmatrix} (G_1 + G_2) & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

Calculate the node voltages



Apply KCL at Node $\rightarrow$ ①

$$I_1 = I_2 + I_3$$

$$5 = \frac{V_1 - V_2}{4} + \frac{V_1}{2}$$

$$20 = V_1 - V_2 + 2V_1$$

$$\text{OR, } \boxed{3V_1 - V_2 = 20}$$

Apply KCL at Node $\rightarrow$ ②

$$I_2 + I_4 = I_5 + I_1$$

$$\frac{V_1 - V_2}{4} + 10 = 5 + \frac{V_2}{6}$$

$$\boxed{-3V_1 + 5V_2 = 60}$$

$$V_1 = \frac{40}{3} = 13.33 \text{ V}$$

$$V_2 = 20 \text{ V}$$

(OR)

$$GV = i$$

$$\begin{bmatrix} \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{6} + \frac{1}{4} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ \frac{10}{-5} \end{bmatrix} = 5$$

$$\frac{3}{4}V_1 - \frac{1}{4}V_2 = 5$$

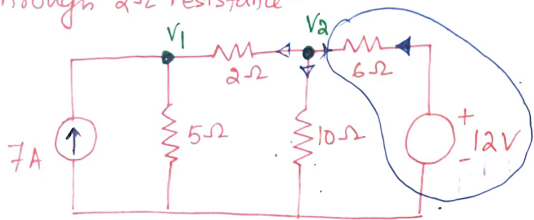
$$\boxed{(3V_1 - V_2 = 20)} \quad \checkmark$$

$$-\frac{V_1}{4} + \frac{5V_2}{12} = 5$$

$$\frac{-3V_1 + 5V_2}{12} = 5$$

$$\boxed{-3V_1 + 5V_2 = 60}$$

Q. Determine the nodal voltages and current through 2Ω resistance -



$$GV = I$$

$$\begin{bmatrix} \frac{1}{5} + \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} + \frac{1}{10} + \frac{1}{6} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 7A \\ \frac{12A}{6} \end{bmatrix}$$

$$\left(\frac{1}{5} + \frac{1}{2}\right)V_1 - \frac{V_2}{2} = 7 \Rightarrow 0.7V_1 - 0.5V_2 = 7$$

$$\Rightarrow -\frac{1}{2}V_1 + \frac{23}{30}V_2 = \frac{12-V_2}{6}$$

$$\Rightarrow \frac{-1}{2}V_1 + \frac{23}{20}V_2 + \frac{V_2}{6} = 2$$

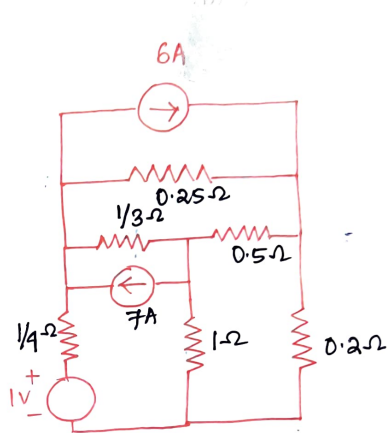
$$\Rightarrow \frac{-V_1}{2} + \frac{79}{60} V_2 = 2$$

$$\begin{cases} V_1 = 22.2 \text{ V} \\ V_2 = 17.1 \text{ V} \end{cases}$$

$$I_{12} = \frac{(22 \cdot 2 - 17 \cdot 1)}{2}$$

$$= 2.55 \text{ A}$$

Qo on the ckt shown (a) find all the node voltages and the current in resistances $0.25\text{-}\Omega$ and $1/3\text{-}\Omega$ use the Nodal Method.



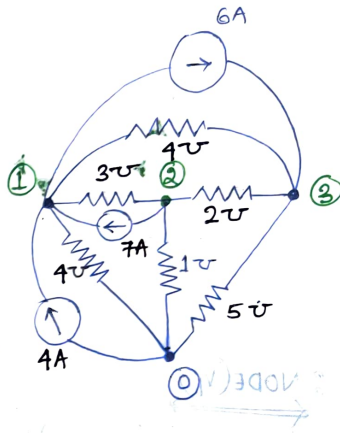
$$[G][v] = [i]$$

$$\begin{bmatrix} 11 & -3 & -4 \\ -3 & 6 & -2 \\ -4 & -2 & 11 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -7 \\ 6 \end{bmatrix}$$

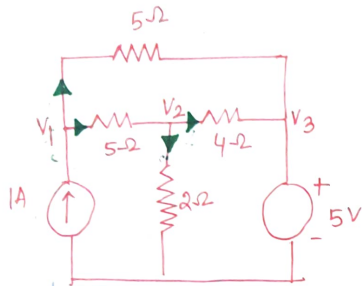
$$\begin{cases} V_1 = 0.462V \\ V_2 = -0.743V \\ V_3 = -0.579V \end{cases}$$

$$\begin{aligned} i_{13}(0.25s) &= \frac{v_1 - v_3}{0.25} \\ &= 4(v_1 - v_3) \\ &= -0.468 \text{ A} \end{aligned}$$

$$i_{12}(1/3-2) = \underline{\underline{3.615 \text{ A}}}$$



Q. for the circuit find the value of V_1 and V_2
 Also find the power input/output of the
 current source and voltage source.



⊙ NODE (V_1)

$$-1 + \left(\frac{V_1 - V_3}{5} \right) + \frac{V_1 - V_2}{5} = 0$$

$$-1 + \left\{ \frac{V_1 - 5}{5} \right\} + \frac{V_1 - V_2}{5} = 0$$

$$-1 + \left(\frac{1}{5} + \frac{1}{5} \right) V_1 - \frac{V_2}{5} - 1 = 0$$

$$\boxed{2V_1 - V_2 = 10}$$

$$-\left(\frac{V_1 - V_2}{5} \right) + \frac{V_2}{2} + \frac{V_2 - V_3}{4} = 0$$

$$-\frac{V_1}{5} + \frac{V_2}{5} + \frac{V_2}{2} + \frac{V_2 - 5}{4} = 0$$

$$\Rightarrow -\frac{V_1}{5} + \left(\frac{1}{2} + \frac{1}{5} + \frac{1}{4} \right) V_2 - \frac{5}{4} = 0$$

$$\boxed{-4V_1 + 19V_2 = 25}$$

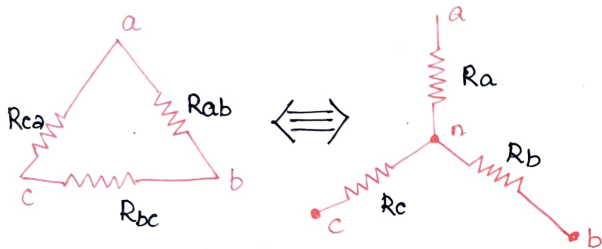
$$V_1 = 6.325 \text{ V}$$

$$V_2 = 2.65 \text{ V}$$

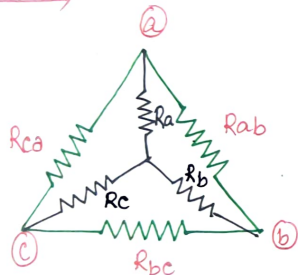
$$\text{current source o/p} = 1 \times 6.325 = 6.325 \text{ W}$$

$$\text{current output of voltage source} =$$

STAR(Y) - DELTA(Δ) CONVERSION



Δ-Y



$$R_a = \frac{R_{ab} \cdot R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_b = \frac{R_{ab} \cdot R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{ca} \cdot R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

Y-Δ

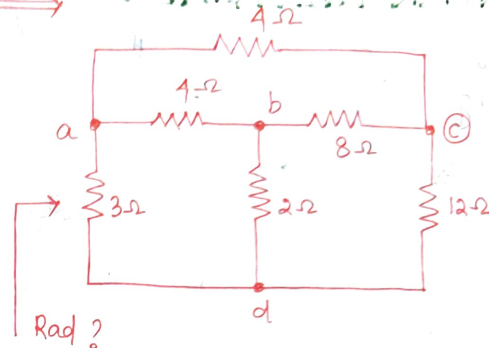
$$R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}$$

$$R_{bc} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_a}$$

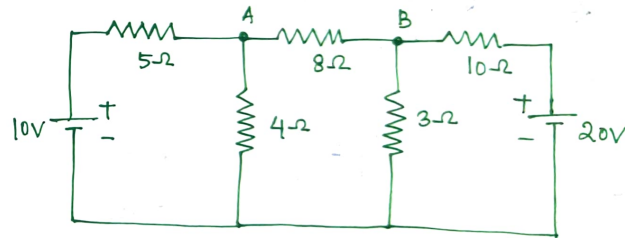
$$R_{ca} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_b}$$

$R_{\Delta} = 3R_Y$
↓
Balanced system.

Solve: Find R_{ad} ?



Q.1: → Apply the node voltage method to find the voltage across A and B; current through 8Ω resistor of the circuit



ANS: $V_{AB} = 0.11V$; $V_1 = 4.47V$ $V_2 = 4.58V$
 $I_{8\Omega} = \frac{V_{AB}}{8} = \frac{0.11}{8} = 0.01375A$ from B to A

NETWORK THEOREMS :→

LINEARITY: A system is linear if it satisfies the following

Homogeneity property :

The homogeneity property requires that if input (excitation) is multiplied by a constant then the output (response) is multiplied by same constant.

$$V = IR$$

$$KIR = KV$$

Additivity property :

$$V_1 = I_1 R$$

$$V_2 = I_2 R$$

$$V = (I_1 + I_2)R = I_1 R + I_2 R \\ = V_1 + V_2$$

(A) SUPERPOSITION THEOREM :

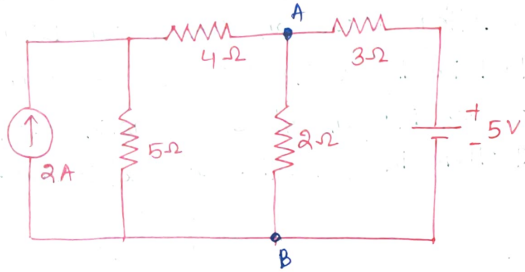
In any linear circuit containing multiple independent sources, the total current through or voltage across an element can be determined algebraically adding the voltage or current due to each independent source acting alone with all other independent sources set to zero.

- Independent v/g source is set to zero
 { short circuit }
- Independent current source is set to zero
 { open circuit }

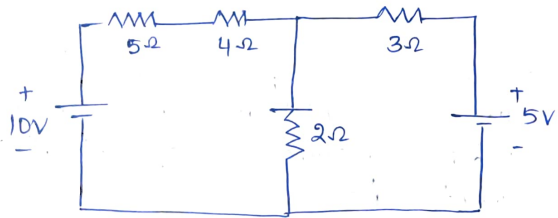
Methodology :

- consider one independent source (either voltage or current) at a time, short ext all other v/g source and open ext all other current sources.
- dependent source can't be set to zero as they are controlled by other circuit parameters
- calculate the current or voltages due to single source using any method (KCL, KVL, Mesh, Nodal }
- Repeat the above steps for each source
- Algebraically add all the results obtained by each source to get the total response.

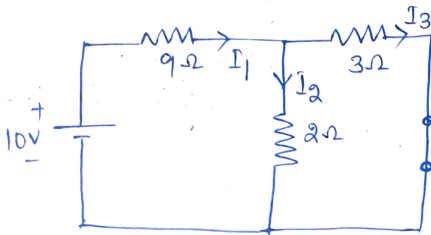
Q. Find the current flowing through 2Ω resistor of the circuit shown by applying Superposition



ANS



STEP: 1: considering $\frac{1}{4}$ source of 10V

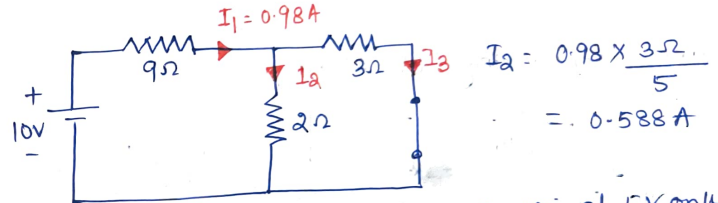


$$R_{eq} = \frac{2 \times 3}{5} + 9$$

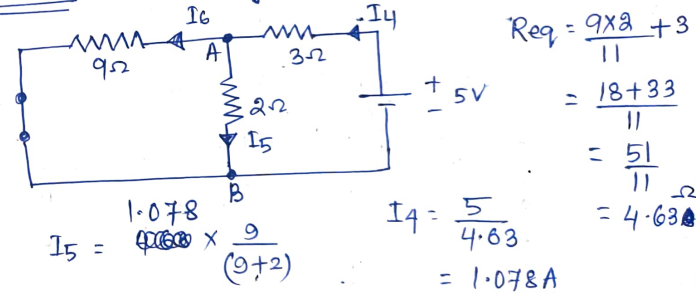
$$= \frac{6}{5} + 9$$

$$= \frac{51}{5} = 10.2\Omega$$

$$I_1 = \frac{10}{10.2} = 0.98A$$



STEP: 2: considering voltage source of 5V only



$$I_5 = \frac{1.078}{(9+2)} \times \frac{9}{5}$$

$$= 0.882A$$

$$I_4 = \frac{5}{4.63} = 1.078A$$

$$R_{eq} = \frac{9 \times 2}{11} + 3$$

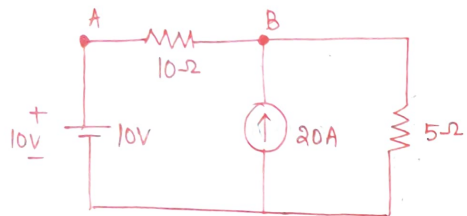
$$= \frac{18+33}{11}$$

$$= \frac{51}{11}$$

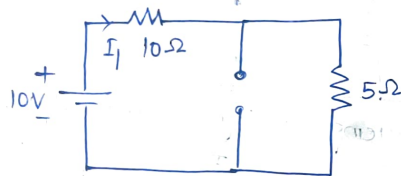
$$= 4.63\Omega$$

$$\text{Total current} = (I_2 + I_5) = 1.47A$$

Q. find the current flowing in the $10\text{-}\Omega$ resistor of the circuit shown by applying superposition principle -

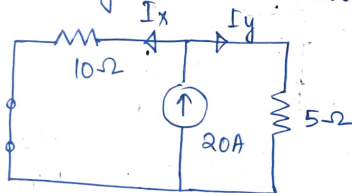


ANS: Considering 10V source



$$I_1 = \frac{10}{15} = 0.666 \text{ A}$$

considering 20A current source



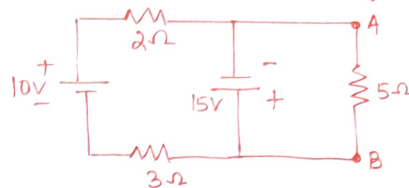
$$I_x = \frac{5 \times 20}{15} = \frac{100}{15} = 6.666 \text{ A}$$

from A to B

$$I_{\text{Total}} = I_1 - I_x = 0.666 - 6.666 = -6 \text{ A}$$

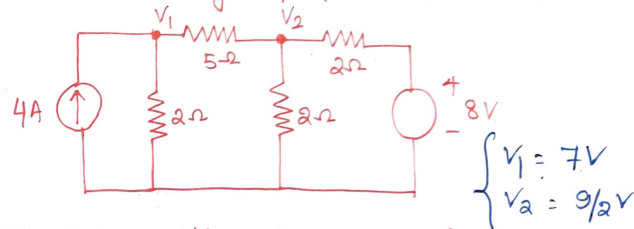
from B to A = 6 A

Q. find the current flowing through $5\text{-}\Omega$ resistor



ANS: 3A from B to A

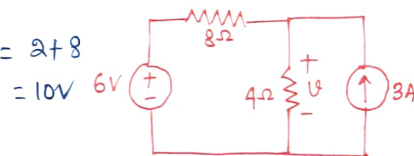
Q. for the ckt shown find the node voltages V_1 and V_2 using superposition theorem -



$$\begin{cases} V_1 = 7\text{V} \\ V_2 = 9/2\text{V} \end{cases}$$

Q. use superposition theorem to find V in the ckt

$$\text{ANS: } V = V_1 + V_2 = 2 + 8 = 10\text{V}$$



Q. using superposition theorem, find V_0 in the circuit

ANS: (7.4A) .

